

RBAC Implementation in Windows

Cybersecurity Portfolio Project

1. Project Overview

This project focused on implementing Role-Based Access Control (RBAC) in Windows using local users, groups, and folder permissions. The lab demonstrated how access rights can be assigned based on user roles to improve security and limit unauthorized access to sensitive resources.

2. Objective

The goal of this project was to create user accounts and security groups, configure role-based permissions, and manage folder access using Windows security settings and advanced permissions.

3. Tools Used

- Windows 10
- Computer Management
- Local Users and Groups
- NTFS Permissions
- Windows Security Settings
- VirtualBox

4. Skills Demonstrated

- Role-Based Access Control (RBAC)
- Identity & Access Management (IAM)
- User Management
- Group Management
- NTFS Permissions
- Access Control
- Windows Administration

5. Lab Environment

The lab environment was created using a Windows 10 virtual machine inside VirtualBox. Local users, groups, and folder security settings were configured using Windows administrative tools.

6. Steps Performed

1. Opened Computer Management.
2. Created a new user account named “mary.”
3. Navigated to Local Users and Groups.
4. Created and reviewed security groups.
5. Added users to the Sales security group.
6. Created and configured secured folders.
7. Reviewed NTFS folder permissions.
8. Opened Advanced Security Settings.
9. Reviewed inherited permissions and access entries.
10. Examined role-based access configurations for users and groups.

7. Screenshots

Figure 1 — Creating a New User Account

- Created a new Windows user account named “mary.”
- Configured password and account settings.

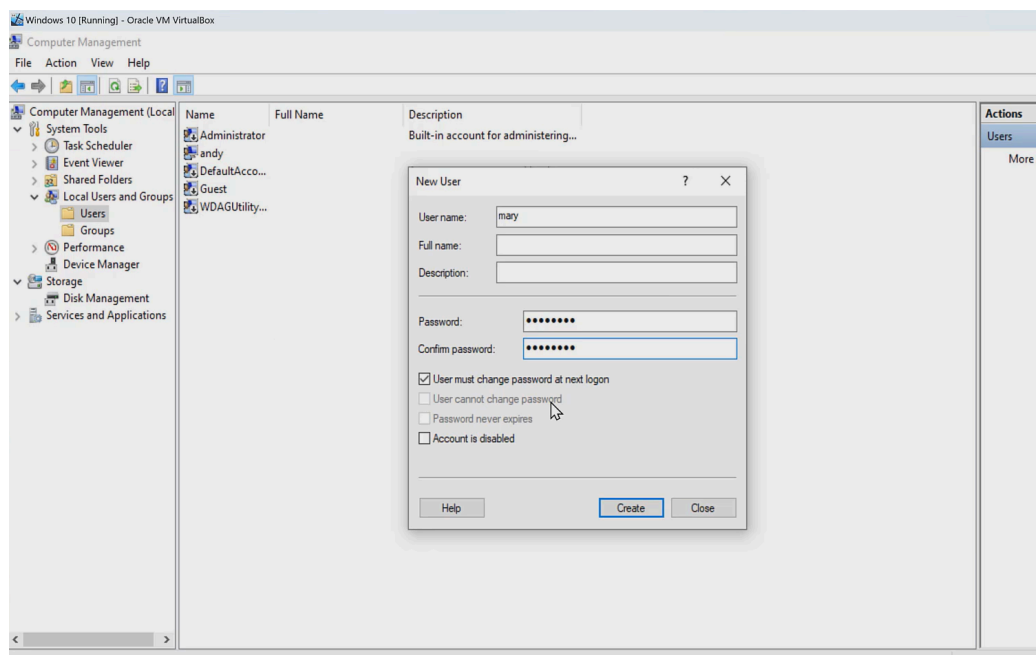


Figure 2 — Managing Security Groups

- Opened Local Users and Groups.
- Reviewed the “Sales” security group.
- Verified group membership configuration.

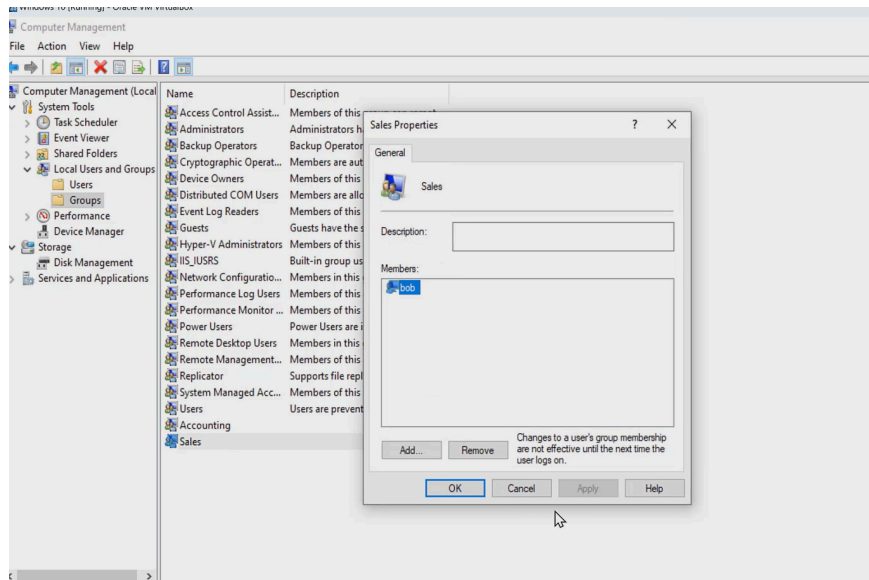


Figure 3 — Reviewing Folder Security Permissions

- Opened NTFS security settings for the “Accounting” folder.
- Reviewed group and user access permissions.

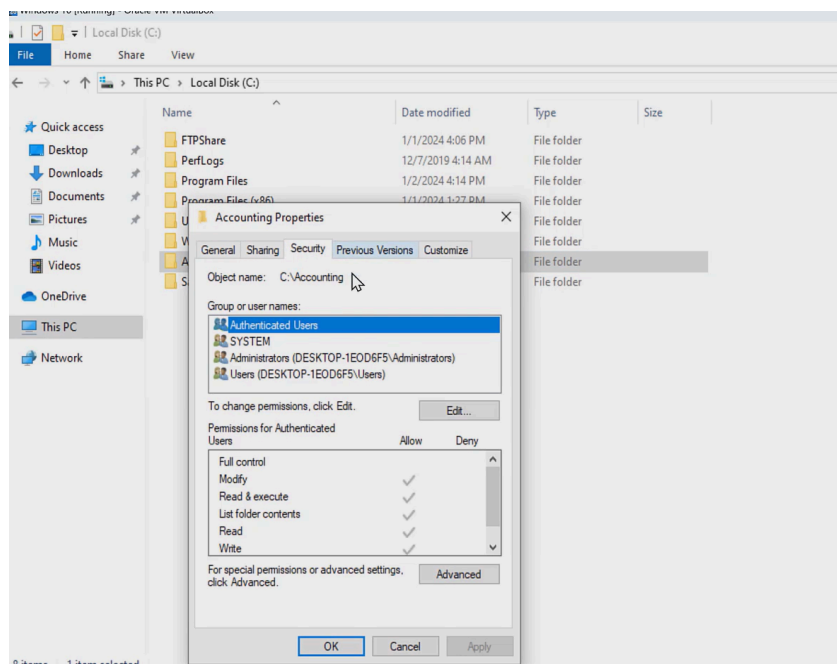
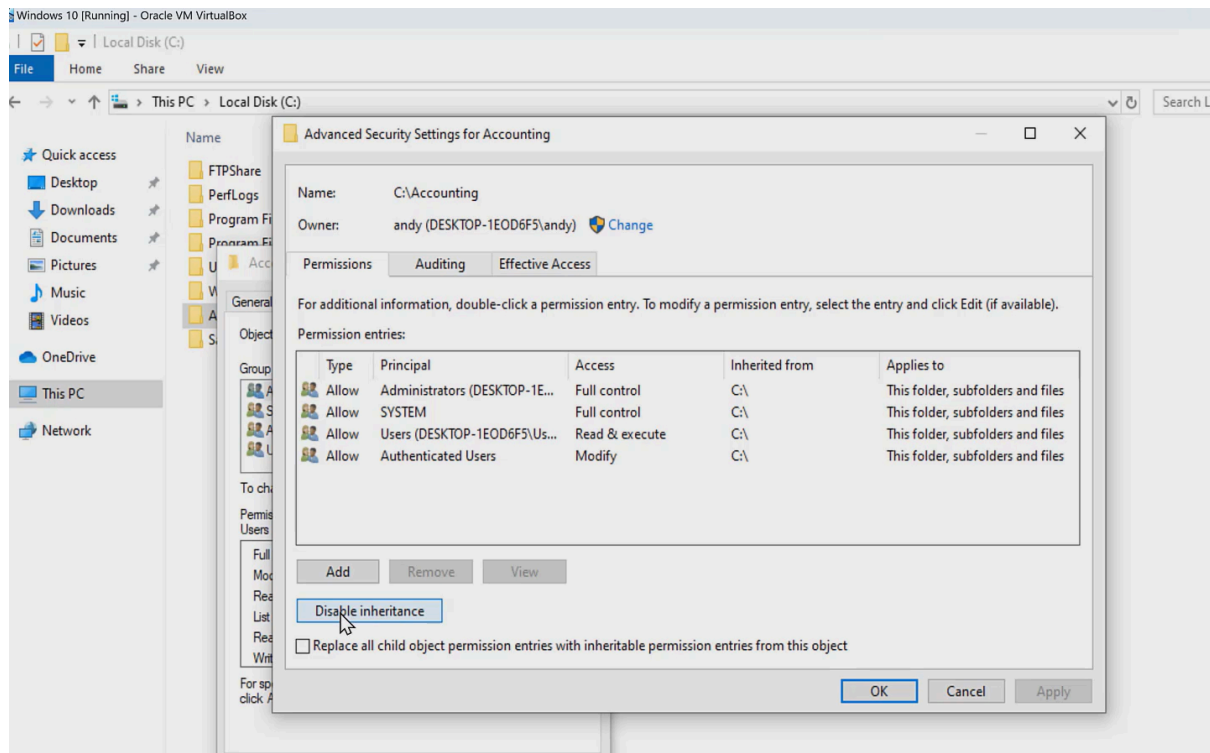


Figure 4 — Advanced Security Settings

- Reviewed advanced NTFS permission entries.
- Examined inherited permissions and effective access settings.



8. Security Benefits

RBAC improves security by assigning permissions based on job roles rather than individual users.

RBAC helps organizations:

- Reduce unauthorized access
- Enforce least privilege
- Simplify permission management
- Improve access control
- Protect sensitive data

9. What I Learned

I learned how Windows uses groups and NTFS permissions to implement role-based access control and manage access to secured resources.

I also gained experience configuring user accounts, security groups, and folder permissions in a Windows environment.

10. Conclusion

This project demonstrated practical cybersecurity skills related to identity and access management, role-based access control, and Windows permission management using local users, groups, and NTFS security settings.